No.	Status	Date
HIS-2006-0412-DS-6-RA-103	Issued	2006-02-24

UT Thickness Data Sheet - 6-RA-103

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 6-RA-103, a rich amine header located within the ATU-100. The scope of work is defined by the 10-year full-unit survey detailed in HIS-2006-0412.

Kevin Doucette, acting as the lead UT technician, is responsible for the execution of the measurements and the recording of the data. All activities were performed in accordance with the site NDE procedure QSCP-NDT-001 Rev 0 and the HIS-UT-002 Rev 1 procedure submitted for QSCP acceptance.

Technical Specifications

Line 6-RA-103 is constructed from A106 Gr.B with an outer diameter of 168.3 mm and a nominal thickness of 10.97 mm. The design limits for this line are 862 kPa(g) and 82 ?C.

The minimum allowable thickness (t-min) is 4.57 mm. Any reading recorded below this retirement thickness shall be raised as an NCR.

Inspection Methodology

Thickness measurements were conducted using ultrasonic testing in compliance with ASTM E797. The methodology follows the requirements for in-service NDE as specified for the ATU-100 unit.

Data acquisition and probe application were managed per the standards established in HIS-UT-002 Rev 1 and the overarching site requirements of QSCP-NDT-001 Rev 0.

Evaluation and Scheduling

The remaining life of 6-RA-103 is calculated as (t - t-min) / rate, where the rate is the larger of the long-term or short-term corrosion rate. The next thickness reading is due at no more than half the remaining life.

As a Class 2 line, the maximum interval for thickness measurements is 10 years. This survey, documented in HIS-2006-0412, fulfills the Class 2 requirement for the 10-year cycle.

Thickness readings, measured 2006-02-24

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2001-03-02	6.82	6.59	0.053	0.046	>50	2011-02-24	
6-SG-101-02	7.11	3.56	2001-03-02	6.71	6.51	0.054	0.040	>50	2011-02-24	
6-SG-101-03	7.11	3.56	2001-03-02	6.74	6.42	0.066	0.064	43.2	2011-02-24	
6-SG-101-04	7.11	3.56	2001-03-02	6.73	6.28	0.081	0.090	30.1	2011-02-24	
6-SG-101-05	7.11	3.56	2001-03-02	6.68	6.22	0.092	0.092	28.8	2011-02-24	
6-SG-101-06	7.11	3.56	2001-03-02	6.87	6.49	0.058	0.076	38.4	2011-02-24	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2001-03-02	6.72	6.26	0.088	0.092	29.2	2011-02-24	
6-SG-102-02	7.11	3.56	2001-03-02	6.72	6.25	0.086	0.094	28.5	2011-02-24	
6-SG-102-03	7.11	3.56	2001-03-02	6.69	6.26	0.078	0.086	31.3	2011-02-24	
6-SG-102-04	7.11	3.56	2001-03-02	6.64	6.22	0.077	0.084	31.6	2011-02-24	
6-SG-102-05	7.11	3.56	2001-03-02	6.93	6.72	0.041	0.042	>50	2011-02-24	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	1996-03-08	8.20	7.89	0.031	0.031	>50	2016-02-24	
8-TG-101-02	8.18	4.06	1996-03-08	8.20	7.75	0.045	0.045	>50	2016-02-24	
8-TG-101-03	8.18	4.06	1996-03-08	8.23	7.79	0.044	0.044	>50	2016-02-24	
8-TG-101-04	8.18	4.06	1996-03-08	8.10	7.76	0.034	0.034	>50	2016-02-24	
8-TG-101-05	8.18	4.06	1996-03-08	8.22	7.95	0.027	0.027	>50	2016-02-24	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	1996-03-08	5.99	5.51	0.048	0.048	48.4	2016-02-24	
4-LA-101-02	6.02	3.18	1996-03-08	6.01	5.76	0.025	0.025	>50	2016-02-24	
4-LA-101-03	6.02	3.18	1996-03-08	5.96	5.76	0.020	0.020	>50	2016-02-24	
4-LA-101-04	6.02	3.18	1996-03-08	6.07	5.78	0.029	0.029	>50	2016-02-24	
4-LA-101-05	6.02	3.18	1996-03-08	6.01	5.81	0.020	0.020	>50	2016-02-24	
4-LA-101-06	6.02	3.18	1996-03-08	6.06	5.64	0.042	0.042	>50	2016-02-24	
4-LA-101-07	6.02	3.18	1996-03-08	5.99	5.57	0.042	0.042	>50	2016-02-24	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	1996-03-08	6.02	5.63	0.039	0.039	>50	2016-02-24	
4-LA-102-02	6.02	3.18	1996-03-08	6.07	5.67	0.040	0.040	>50	2016-02-24	
4-LA-102-03	6.02	3.18	1996-03-08	6.07	5.55	0.052	0.052	45.4	2016-02-24	
4-LA-102-04	6.02	3.18	1996-03-08	5.99	5.34	0.065	0.065	33.1	2016-02-24	
4-LA-102-05	6.02	3.18	1996-03-08	6.00	5.62	0.038	0.038	>50	2016-02-24	
4-LA-102-06	6.02	3.18	1996-03-08	6.00	5.83	0.017	0.017	>50	2016-02-24	
4-LA-102-07	6.02	3.18	1996-03-08	6.00	5.63	0.037	0.037	>50	2016-02-24	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	1996-03-08	7.02	6.75	0.027	0.027	>50	2016-02-24	
6-LA-103-02	7.11	3.56	1996-03-08	7.19	6.87	0.032	0.032	>50	2016-02-24	
6-LA-103-03	7.11	3.56	1996-03-08	7.08	6.65	0.043	0.043	>50	2016-02-24	
6-LA-103-04	7.11	3.56	1996-03-08	7.17	6.52	0.065	0.065	45.4	2016-02-24	
6-LA-103-05	7.11	3.56	1996-03-08	7.19	6.49	0.070	0.070	41.7	2016-02-24	
6-LA-103-06	7.11	3.56	1996-03-08	7.16	6.42	0.074	0.074	38.5	2016-02-24	
6-LA-103-07	7.11	3.56	1996-03-08	7.18	6.76	0.042	0.042	>50	2016-02-24	
6-LA-103-08	7.11	3.56	1996-03-08	7.04	6.79	0.025	0.025	>50	2016-02-24	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2001-03-02	8.39	8.27	0.025	0.024	>50	2016-02-24	
4-RA-101-02	8.56	3.81	2001-03-02	8.45	8.27	0.027	0.036	>50	2016-02-24	
4-RA-101-03	8.56	3.81	2001-03-02	8.42	8.10	0.051	0.064	>50	2016-02-24	
4-RA-101-04	8.56	3.81	2001-03-02	8.37	8.34	0.029	0.006	>50	2016-02-24	
4-RA-101-05	8.56	3.81	2001-03-02	8.42	8.29	0.031	0.026	>50	2016-02-24	
4-RA-101-07	8.56	3.81	2001-03-02	8.51	8.30	0.030	0.042	>50	2016-02-24	
4-RA-101-08	8.56	3.81	2001-03-02	8.30	8.07	0.061	0.046	>50	2016-02-24	
4-RA-101-09	8.56	3.81	2001-03-02	8.43	8.43	0.014	0.000	>50	2016-02-24	
4-RA-101-10	8.56	3.81	2001-03-02	8.48	8.27	0.040	0.042	>50	2016-02-24	
4-RA-101-11	8.56	3.81	2001-03-02	8.39	8.14	0.054	0.050	>50	2016-02-24	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2001-03-02	8.48	8.41	0.013	0.014	>50	2016-02-24	
4-RA-102-02	8.56	3.81	2001-03-02	8.50	8.15	0.042	0.070	>50	2016-02-24	
4-RA-102-03	8.56	3.81	2001-03-02	8.37	8.27	0.029	0.020	>50	2016-02-24	
4-RA-102-04	8.56	3.81	2001-03-02	8.35	8.24	0.030	0.022	>50	2016-02-24	
4-RA-102-05	8.56	3.81	2001-03-02	8.36	8.19	0.036	0.034	>50	2016-02-24	
4-RA-102-06	8.56	3.81	2001-03-02	8.38	8.15	0.036	0.046	>50	2016-02-24	
4-RA-102-07	8.56	3.81	2001-03-02	8.49	8.29	0.029	0.040	>50	2016-02-24	
4-RA-102-08	8.56	3.81	2001-03-02	8.36	8.05	0.044	0.062	>50	2016-02-24	
4-RA-102-09	8.56	3.81	2001-03-02	8.35	8.18	0.037	0.034	>50	2016-02-24	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2001-03-02	10.82	10.76	0.024	0.012	>50	2016-02-24	
6-RA-103-02	10.97	4.57	2001-03-02	10.69	10.50	0.059	0.038	>50	2016-02-24	
6-RA-103-03	10.97	4.57	2001-03-02	10.84	10.67	0.032	0.034	>50	2016-02-24	
6-RA-103-04	10.97	4.57	2001-03-02	10.59	10.40	0.058	0.038	>50	2016-02-24	
6-RA-103-05	10.97	4.57	2001-03-02	10.61	10.34	0.061	0.054	>50	2016-02-24	
6-RA-103-06	10.97	4.57	2001-03-02	10.86	10.74	0.027	0.024	>50	2016-02-24	
6-RA-103-07	10.97	4.57	2001-03-02	10.79	10.66	0.033	0.026	>50	2016-02-24	
6-RA-103-08	10.97	4.57	2001-03-02	10.83	10.58	0.039	0.050	>50	2016-02-24	
6-RA-103-09	10.97	4.57	2001-03-02	10.85	10.67	0.034	0.036	>50	2016-02-24	
6-RA-103-10	10.97	4.57	2001-03-02	10.78	10.27	0.075	0.102	>50	2016-02-24	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2001-03-02	7.26	6.98	0.061	0.056	>50	2011-02-24	
3-RV-101-02	7.62	3.56	2001-03-02	7.31	6.93	0.064	0.076	44.2	2011-02-24	
3-RV-101-03	7.62	3.56	2001-03-02	7.15	6.57	0.095	0.116	25.9	2011-02-24	
3-RV-101-04	7.62	3.56	2001-03-02	7.18	6.65	0.094	0.106	29.1	2011-02-24	
3-RV-101-05	7.62	3.56	2001-03-02	7.25	7.01	0.066	0.048	>50	2011-02-24	
3-RV-101-06	7.62	3.56	2001-03-02	7.20	6.74	0.085	0.092	34.4	2011-02-24	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2001-03-02	5.29	4.85	0.068	0.088	23.3	2011-02-24	
2-RV-102-02	5.54	2.79	2001-03-02	5.12	4.84	0.075	0.056	27.2	2011-02-24	
2-RV-102-03	5.54	2.79	2001-03-02	5.24	5.14	0.034	0.020	>50	2011-02-24	
2-RV-102-04	5.54	2.79	2001-03-02	5.33	5.07	0.051	0.052	43.7	2011-02-24	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	1996-03-08	5.48	5.48	0.000	0.000	>50	2016-02-24	
3-RF-101-02	5.49	2.41	1996-03-08	5.45	5.37	0.008	0.008	>50	2016-02-24	
3-RF-101-03	5.49	2.41	1996-03-08	5.45	5.45	0.000	0.000	>50	2016-02-24	
3-RF-101-04	5.49	2.41	1996-03-08	5.46	5.33	0.013	0.013	>50	2016-02-24	
3-RF-101-05	5.49	2.41	1996-03-08	5.45	5.40	0.005	0.005	>50	2016-02-24	
3-RF-101-06	5.49	2.41	1996-03-08	5.52	5.40	0.012	0.012	>50	2016-02-24	
3-RF-101-07	5.49	2.41	1996-03-08	5.42	5.35	0.007	0.007	>50	2016-02-24	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	1996-03-08	3.92	3.85	0.007	0.007	>50	2016-02-24	
2-RF-102-02	3.91	2.16	1996-03-08	3.88	3.88	0.000	0.000	>50	2016-02-24	
2-RF-102-03	3.91	2.16	1996-03-08	3.93	3.80	0.013	0.013	>50	2016-02-24	
2-RF-102-04	3.91	2.16	1996-03-08	3.87	3.81	0.006	0.006	>50	2016-02-24	
2-RF-102-07	3.91	2.16	1996-03-08	3.96	3.85	0.011	0.011	>50	2016-02-24	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	1996-03-08	6.07	5.75	0.032	0.032	>50	2016-02-24	
4-LA-104-02	6.02	3.18	1996-03-08	6.03	5.56	0.047	0.047	>50	2016-02-24	
4-LA-104-03	6.02	3.18	1996-03-08	5.96	5.74	0.022	0.022	>50	2016-02-24	
4-LA-104-04	6.02	3.18	1996-03-08	6.06	5.86	0.020	0.020	>50	2016-02-24	
4-LA-104-05	6.02	3.18	1996-03-08	6.04	5.51	0.053	0.053	43.8	2016-02-24	
4-LA-104-06	6.02	3.18	1996-03-08	6.10	5.83	0.027	0.027	>50	2016-02-24	
4-LA-104-07	6.02	3.18	1996-03-08	6.01	5.62	0.039	0.039	>50	2016-02-24	
4-LA-104-08	6.02	3.18	1996-03-08	5.94	5.50	0.044	0.044	>50	2016-02-24	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2001-03-02	8.40	8.16	0.035	0.048	>50	2016-02-24	
4-RA-104-02	8.56	3.81	2001-03-02	8.41	8.23	0.032	0.036	>50	2016-02-24	
4-RA-104-03	8.56	3.81	2001-03-02	8.46	8.30	0.024	0.032	>50	2016-02-24	
4-RA-104-04	8.56	3.81	2001-03-02	8.35	8.21	0.038	0.028	>50	2016-02-24	
4-RA-104-05	8.56	3.81	2001-03-02	8.29	8.14	0.046	0.030	>50	2016-02-24	
4-RA-104-06	8.56	3.81	2001-03-02	8.39	8.09	0.050	0.060	>50	2016-02-24	
4-RA-104-07	8.56	3.81	2001-03-02	8.29	7.97	0.062	0.064	>50	2016-02-24	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	1996-03-08	3.93	3.64	0.029	0.029	46.4	2016-02-24	
2-ST-101-02	3.91	2.29	1996-03-08	3.90	3.67	0.023	0.023	>50	2016-02-24	
2-ST-101-03	3.91	2.29	1996-03-08	3.94	3.82	0.012	0.012	>50	2016-02-24	
2-ST-101-04	3.91	2.29	1996-03-08	3.82	3.77	0.005	0.005	>50	2016-02-24	
2-ST-101-05	3.91	2.29	1996-03-08	3.91	3.72	0.019	0.019	>50	2016-02-24	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	1996-03-08	5.42	5.13	0.029	0.029	>50	2016-02-24	
3-CW-101-02	5.49	2.79	1996-03-08	5.41	5.20	0.021	0.021	>50	2016-02-24	
3-CW-101-03	5.49	2.79	1996-03-08	5.53	5.08	0.045	0.045	>50	2016-02-24	
3-CW-101-04	5.49	2.79	1996-03-08	5.46	5.38	0.008	0.008	>50	2016-02-24	
3-CW-101-05	5.49	2.79	1996-03-08	5.48	5.33	0.015	0.015	>50	2016-02-24	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	1996-03-08	5.50	5.20	0.030	0.030	>50	2016-02-24	
3-CW-102-02	5.49	2.79	1996-03-08	5.51	5.02	0.049	0.049	45.4	2016-02-24	
3-CW-102-03	5.49	2.79	1996-03-08	5.53	5.07	0.046	0.046	49.4	2016-02-24	
3-CW-102-04	5.49	2.79	1996-03-08	5.51	5.03	0.048	0.048	46.5	2016-02-24	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2006-02-24	_____